

Applied Data Science

Phase 2: Machine Learning for Greek Legal Texts

Supervised and Unsupervised Machine Learning

May 2025

Maria Schoinaki - 3210191

Dataset

We use the **Greek Legal Code** dataset provided by the **University of Athens** (*AI-team-UoA/greek_legal_code*), which contains approximately **47,000 legal documents**. Each document is **annotated** with **three types** of labels: **volume (47 classes)**, **chapter (389 classes)**, and **subject (2285 classes)**. The dataset is divided into **training**, **validation**, and **test** sets for each classification level.

B1. In this section, we apply **supervised machine learning** methods to **Greek legal texts**, focusing on the classification of **legal documents** at **three different categorical levels**. Our goal is to **predict the correct category** (*volume, chapter, or subject*) for each document using a variety of **machine learning models** and **text representations**.

It is worth noting that **higher scores** could **potentially** be **achieved** by applying more **sophisticated** text **preprocessing** (*e.g., stemming or lemmatization for Greek*), using **additional features**, or **employing more computationally demanding models**. However, these approaches would **require considerably more time** and **computational resources**, which were beyond the scope and constraints of the **present assignment**.

For each **classification level**, we trained the models and selected the **best regularization parameter (C)** using **grid search** on the **validation set**. The grid search was conducted over **four C values** (**0.01, 0.1, 1, 10**), which represent a **logarithmic range** from **strong** to **weak regularization**, thus **providing a good trade-off** between **model performance** and **computational efficiency**.

I. SVM Training and Evaluation

- An automated pipeline is defined for training and evaluating a **Support Vector Machine (SVM)** classifier for text classification.
- Both **TF-IDF** and **Bag-of-Words (BoW)** representations are considered.
- The function **applies grid search** on the **SVM's** regularization parameter (**C**) to select the best model based on **weighted F1-score** on the validation set.
- Performance metrics (**accuracy, precision, recall, F1-score**) are computed using **weighted averages** to address **class imbalance**.

This approach is applied to **all classification levels** for **robust comparison**.

II. Logistic Regression (LR) with Word Embeddings

- Used **Logistic Regression** classifier with **two types** of **word embeddings**:
- **Word2Vec**: Documents represented as the average of their word embeddings, **leveraging pre-trained vectors**.
- **FastText**: Similar **approach** but using **FastText embeddings** for richer **subword** information.
- **Evaluated** performance for **each classification level**.

III. Naive Bayes (NB)

- Implemented **Multinomial Naive Bayes** classifier using **TF-IDF** features.
- Focused on **efficiency** and **scalability** for **text classification** tasks.

B2. In this section, we apply **unsupervised machine learning** methods to **Greek legal texts**, focusing on the classification of **legal documents** of the **Supreme Court of Greece**.

This analysis explores a dataset of **Greek Supreme Court legal** decisions, focusing on both **exploratory data analysis (EDA)** and **unsupervised learning (clustering)**. The aim is to uncover **patterns** in **legal categories** and **tags**, understand class **imbalances**, and use **K-means clustering** (*with large language models for cluster interpretation*) to surface **thematic groupings**.

1. Exploratory Data Analysis (EDA)

Top Categories & Tags:

- The majority of cases belong to a handful of dominant categories and tags (mainly procedural/reasoning issues like “Sufficiency of Reasoning” and “Absolute Nullity”), showing strong class imbalance.
- Both categories and tags exhibit a “long tail” distribution, with most cases having 1–4 tags but some much more, confirming the multilabel nature of the data.

Co-occurrence & Complexity:

- Many cases have several tags, and key categories (like procedural issues) tend to co-occur with multiple tags.
- Boxplots and violin plots reveal variability and outliers in tag counts per category, reflecting case complexity.

Associations & Diversity:

- Heatmaps illustrate strong associations between certain categories and tags, while some tags are broadly distributed across many categories (signaling legal generality).Length & Content:
- Summaries for certain categories are longer and more variable, indicating legal complexity.

Statistical Challenges:

- Pareto plots and co-occurrence heatmaps emphasize the challenge of modeling rare categories and multilabel dependencies in supervised or unsupervised tasks.

2. K-means Clustering on Decision Summaries

- **Vectorization:**
Summaries are transformed to TF-IDF vectors (max 3,000 features).
- **Clustering & Evaluation:**
K-means is run with K from 2 to 20.
Silhouette Score: Measures how well-separated the clusters are (peaks at K=18).
 - **NMI (Normalized Mutual Information):** Assesses alignment of clusters with known categories and tags (best at K=20).
- **Visualization:**
 - Line plots show metric trends across K.
 - PCA is used to visualize cluster separation (K=18 or 19).
- **Interpretation:**
 - Higher K yields more alignment with metadata but not sharply distinct clusters (typical for text data).
 - Clusters capture “fuzzy” groupings and legal subtopics.

3. LLM-based Cluster Theme Extraction

- **LLM Summarization:**

For each cluster, a Greek Large Language Model (Llama-Krikri-8B-Instruct via Unsloth) is prompted to generate a theme phrase ("Θέμα: ...") summarizing the core topic of the cluster, using representative summaries (random and centroid-based).

- **Qualitative Cluster Review:**

Thematic titles are reviewed for all clusters. The main legal categories per cluster are presented (via barplots, pie charts, wordclouds).